📦 The "Autonomous Box" Assembly Manual

Project: MacBook-Carrying AI Robot (ESP32-S3 / 4WD)

🛠️ Phase 1: The Motor Prep (Soldering)

Before you build the frame, you must prep the "muscles."

1. **Tin the Tip:** Turn on your iron with the **Chisel Tip**. Once hot, melt a small amount of solder onto the tip until it's shiny.
2. **Flux the Tabs:** Use your **Flux Pen** to "paint" the two metal tabs on all 4 motors.
3. **The "Top Tab" Rule:** * Lay the motor flat with the white axle pointing at you.
    - Solder the **RED** wire to the top tab.
    - Solder the **BLACK** wire to the bottom tab.
    - **Repeat for all 4 motors.** (This ensures they all spin the same way).

🏗️ Phase 2: Chassis Assembly

1. **Peel the Film:** If your chassis plates have brown paper on them, peel it off now to reveal the black/clear surface.
2. **Mount Motors:** Use the long bolts and the small "T-shaped" plastic brackets to sandwich the motors onto the **bottom** plate.
3. **Add Wheels:** Press the yellow wheels onto the white motor axles.
4. **Stand-offs:** Screw the brass hexagonal "pillars" into the corners of the bottom plate. These will hold the "roof" (top plate) later.

⚡ Phase 3: The Nervous System (Wiring)

Do not plug in the batteries until this is finished!

1. The Motor Driver (L298N - The Red Board)

- **Left Motors:** Twist the two Red wires from the left side together and put them in the **OUT1** terminal. Twist the two Black wires together and put them in **OUT2**. Tighten the screws.
- **Right Motors:** Twist the two Red wires from the right side together into **OUT3**. Twist the two Black wires together into **OUT4**. Tighten the screws.

2. Logic Jumper Map (Brain to Muscles)

Use your Female-to-Male jumpers.

ESP32-S3 Pin	Driver Pin	Function
GPIO 1	**IN1**	Left Forward
GPIO 2	**IN2**	Left Backward
GPIO 41	**IN3**	Right Forward
GPIO 42	**IN4**	Right Backward

3. The Power Bridge (CRITICAL)

- Connect the **Red** wire from your battery box to the **+12V** terminal on the L298N.
- Connect the **Black** wire from your battery box to the **GND** terminal on the L298N.
- **The Shared Ground:** Take a jumper wire. Connect one end to a **GND** pin on the ESP32 and the other end to that same **GND** terminal on the L298N.

🐱 Phase 4: Senses (Optional Add-ons)

- **Sonar (HC-SR04):** * VCC → 5V (on ESP32 or Driver)
    - GND → GND
    - Trig → GPIO 5
    - Echo → GPIO 6

🚀 Phase 5: Final Setup

1. **Mount the Mac:** Place your MacBook on the top plate. Secure the "Box" shell around it.
2. **Connect:** Plug your **PS5 Cable** from the Mac into the **UART Port** of the ESP32-S3.
3. **Power Up:** Flip the switch on the battery box.

⚠️ Pro-Builder Tips:

- **The "Tug Test":** After soldering or screwing in a wire, give it a light tug. If it moves, it will fail when the robot starts vibrating.
- **Cable Management:** Use a couple of rubber bands or zip ties to keep the wires away from the spinning wheels.
- **Grounding:** If the robot acts "crazy" (motors twitching), 99% of the time it’s because the **Shared Ground** wire came loose.